

to raise a dispute and ask for your money back. In such cases, there is very little a game developer can do, since the credit card company rarely listens to them. Moreover, game developers must pay US\$ 25 for every disputed transaction. Because of this, the gaming industry loses millions of dollars in customer support.

Blockchain to the rescue

Blockchain technology has the potential to end all these problems in the gaming world. The games based on blockchain technology will offer solutions for game developers as well as the players. There are infinite possibilities that a blockchain has to offer but, for starters, there are several benefits that no one can ignore. Here are some of them.

True ownership of virtual assets: In blockchain games, the virtual assets of the player will belong to the player and the creators cannot take them away. These virtual assets may be stored in the form of tokens in the player's digital wallet. These tokens can be fungible (replaceable by another identical item) or even non-fungible. With blockchain games, the players will not have to worry about losing their digital assets even if they get banned from the game, if the company shuts down or they quit the game.

Inter-game compatibility: As the characters, items or 'karma' of the games are stored on your blockchain, it will allow you to see your existing characters or items even on a completely unrelated game. It could open an era for a new type of game in which characters from separate game genres could play together in one game. In case you are building your own game, instead of building new characters, you can use assets of an existing game or invite players from the large game communities to instantly jump into your game with the characters they already have. This is one of the coolest features blockchain technology has to offer in the gaming industry.

Uniformity in the virtual assets of different games: The design of blockchain games is fundamentally different from that of traditional games. It does not allow the in-game currency to get generated infinitely. Currently, in some games, while gold coins can be generated by just killing some monster, in others, they might be a scarce resource. Blockchain will end this infinite generation of coins in some games.

No Need for sub-servers: The blockchain technology in the MMORPG will end the need to have different servers or even sub-servers for a different region. Hence there will be no blackout in a particular region due to the failure of local servers.

Cheap and secured trading of virtual assets: Blockchain will allow users to convert their virtual assets into a token, making these free for buying, selling and trading, at the will of the owner. The record of the transaction will be on the blockchain itself, eliminating the threat of fraudulent activities. The game characters, items, etc, can be managed through smart contracts as has been done in Cryptokitties – a blockchain game that has created much hype in the blockchain industry.

Blockchain technology even has the potential to offer micro-transactions involving the cryptocurrency

Cases of early blockchain based gaming

Realising the potential of blockchain technology in the next generation of online games, some developers have already started using it. Successful projects range from blockchain game development platforms to innovative payment gateways to buy games. While some of these blockchain based games are already in the market, some others are in their beta phase and some have raised funding through the process of Initial Coin Offering (ICO). Every project has different aims but all of them together hold much promise to build the future of gaming.

Blockchain games in the market

There are several blockchain based games available in the market. Many of them are built on top of the Ethereum blockchain, with Cryptokitties being the most well known. This game uses ERC 721 tokens which are non-fungible, indivisible and unique. It uses smart contracts for ownership and for the distribution of virtual cats. This is considered as the first non-financial application of blockchain technology.

Other blockchain games that are available in the market are Ethermon, Cryptopets, Cryptopuppies, Etherarmy, etc. Ethereum is not the only blockchain to attract game developers. A blockchain game called Spell of Genesis is built on the Bitcoin blockchain.

Payment systems for games

There are several blockchain based platforms which have been built that offer users the option to sell, buy and promote online games. Some of these are GameCredits, Enjincoin and Refereum. Some gaming corporations like Valve are even allowing users to buy games using Bitcoins.

Platforms for building blockchain based games

Platforms for building blockchain based games promise to offer all the new features that we just discussed. They aim to simplify the ownership, sharing and trading of virtual assets. Such platforms like BitGuild and Chimaera are expected to get launched this year. There are several other platforms like Decentraland and Voxels which are offering a virtual space for blockchain based 3D graphics and games.

From the tokenisation of virtual goods to improving betting and payments, every area of gaming is undergoing experimentation using blockchain technology. While this technology has a lot to offer, there is an issue of speed, especially when we talk about MMORPG. At present, there is a blockchain based MMORPG, but it is still in its beta version. **END** 

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